

EGRE 245 Engineering Programming**Fall 2025**

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Programming Assignment 4: Student Record Manager

1 Overview

In this assignment, you will write a C program that manages student records for a small class. Each record should contain a student's **name**, **ID number**, and **three test scores**. The program will dynamically allocate memory for the records, calculate averages, and allow the user to view or modify data using **pointers and structures**.

2 Objectives

- Practice defining and using structures.
- Use pointers to access and modify structure members.
- Pass structures and pointers to functions.
- Reinforce dynamic memory management using `malloc()` and `free()`.

3 Program Requirements

Your program should:

1. Define a structure for student data:

```
struct Student {  
    char name[30];  
    int id;  
    float test[3];  
    float average;  
};
```

2. Ask the user for the number of students, then dynamically allocate memory.

3. Implement the following functions:

- `void inputStudent(struct Student *s)` – prompts for name, ID, and test scores.

- `void computeAverage(struct Student *s)` – calculates the average score.
 - `void displayStudent(const struct Student *s)` – displays one student's info.
 - `struct Student *findHighestAverage(struct Student *list, int n)` – returns a pointer to the student with the highest average.
4. Display all student data in a formatted table.
 5. Identify and display the student with the highest average.
 6. Free allocated memory before exiting.

4 Sample Output

Your output should match the sample run below.

```
Include header information (name, date, etc.)
```

```
Enter number of students: 3
```

```
--- Enter data for Student 1 ---
```

```
Name: Mark Schwitzerlett
```

```
ID: 123874
```

```
Enter 3 test scores: 98 89 92
```

```
--- Enter data for Student 2 ---
```

```
Name: Henri Jacobi
```

```
ID: 543892
```

```
Enter 3 test scores: 100 97 94
```

```
--- Enter data for Student 3 ---
```

```
Name: Marie Curie
```

```
ID: 765239
```

```
Enter 3 test scores: 99 94 89
```

```
-----  
Name                                ID      Average  
-----  
Mark Schwitzerlett                 123874   93.00  
Henri Jacobi                       543892   97.00  
Marie Curie                        765239   94.00  
-----
```

```
Highest Average: Henri Jacobi (97.00)
```

```
...Program finished with exit code 0
```

```
Press ENTER to exit console.
```

5 Deliverables

Submit a single `.c` file named:

`Assignment_PointersStructs_FirstName_LastName.c`

Your submission must include:

- Proper header comment block
- Function prototypes
- Internal documentation and clear formatting

This assignment is due Friday, November 14, at 11:59 pm.